

Module-VI

Lecture-V

Symbolic Model Checking

1. Introduction

In the previous lecture we have discussed about the CTL model checking algorithm. It is a labelling algorithm and we label the states by the given CTL formula if the CTL formula holds in the state. Also we have seen that the complexity of the algorithm is related to the size of the Kripke structure. For complex system, the size of the FSM is prohibitively big and so the model checking takes huge amount of time. Finite state models of concurrent systems grow exponentially as the number of components of the system increases. For example, an FSM having 'n' state variables will have ' $O(2^n)$ ' finite states, i.e., the relation between number of state variables and number of finite states is exponential. It is known widely that the state space explosion problem in automatic verification has limited its use to small systems. To make the model checking more effective, we need some methods to content the state space explosion problem. One possible way is to represent the state space of the FSM in a compact way so that the model checking becomes effective. In Module 4 of the course, we have discussed about ROBDD (Reduced Ordered Binary Decision Diagram) and with the help of ROBDD we can represent the Boolean expression in canonical form. Also with proper variable ordering, most of the time we get the compact representation of the Boolean expression. Also we have seen that ROBDD can be used to represent the Finite State Transition system and the set of states of transition system.

In model checking algorithm we use ROBDD to represent the Kripke structure and the method that we get is named as Symbolic Model Checking. It says that the states are not enumerated but represented symbolically as a set with the help of ROBDDs, that's why the term symbolic is used.

2. Symbolic Model Checking

Model checking using OBDDs is called symbolic model checking. The term emphasises that individual states are not represented; rather, sets of states are represented symbolically, namely, those which satisfy the formula being checked. The set of states of FSM and the transition system can be represented by ROBDD. It is explained with the help of an example. Consider the Kripke structure of Figure 1.

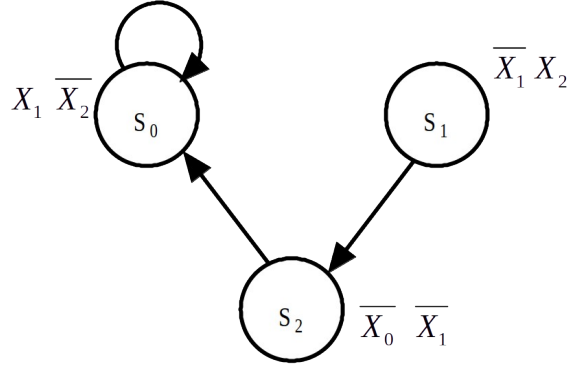


Figure 1: A Simple CTL Model

Given a system, we must assign unique Boolean vectors to each state before we construct OBDD. The easiest way to do this is to assign a Boolean vector based on the propositional atoms true in that state. Here, x_1 and x_2 are propositional atoms.

set of states	representation by boolean values	representation by boolean function
$\emptyset$		0
$\{s_0\}$	(1, 0)	$x_1 \cdot \overline{x_2}$
$\{s_1\}$	(0, 1)	$\overline{x_1} \cdot x_2$
$\{s_2\}$	(0, 0)	$\overline{x_1} \cdot \overline{x_2}$
$\{s_0, s_1\}$	(1, 0), (0, 1)	$x_1 \cdot \overline{x_2} + \overline{x_1} \cdot x_2$
$\{s_0, s_2\}$	(1, 0), (0, 0)	$x_1 \cdot \overline{x_2} + \overline{x_1} \cdot \overline{x_2}$
$\{s_1, s_2\}$	(0, 1), (0, 0)	$\overline{x_1} \cdot x_2 + \overline{x_1} \cdot \overline{x_2}$
S	(1, 0), (0, 1), (0, 0)	$x_1 \cdot \overline{x_2} + \overline{x_1} \cdot x_2 + \overline{x_1} \cdot \overline{x_2}$

Table 1: Representation of Subset of States

Using the Boolean functions presented in the Table 1, we can construct OBDDs for each subset. On the other hand, given a OBDD, it is also possible to identify the subset of states. To represent transition relations, we need two sets of the Boolean

variables. One of the sets must be used to represent the next state. Thus we take primed versions of the Boolean variables to represent next states.

Present State		Next State		
x_1	x_2	x_1'	x_2'	$\rightarrow$
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1

Table 2: Representing Transition Relations

In the Table 2, the link ($\rightarrow$) column is 1 if and only if there exists an edge from present state to next state, i.e., if there is a transition in the system.

The Boolean function representing all transitions of the FSM of Figure 1 is given by,

$$f : \overline{X_1} \overline{X'_1} \overline{X_2} \overline{X'_2} + \overline{X_1} X'_1 \overline{X_2} \overline{X'_2} + \overline{X_1} \overline{X'_1} X_2 \overline{X'_2} + X_1 \overline{X'_1} X_2 \overline{X'_2}$$

The OBDD of this Boolean function is given in Figure 2 with the variable ordering [X_1, X_1', X_2, X_2'].

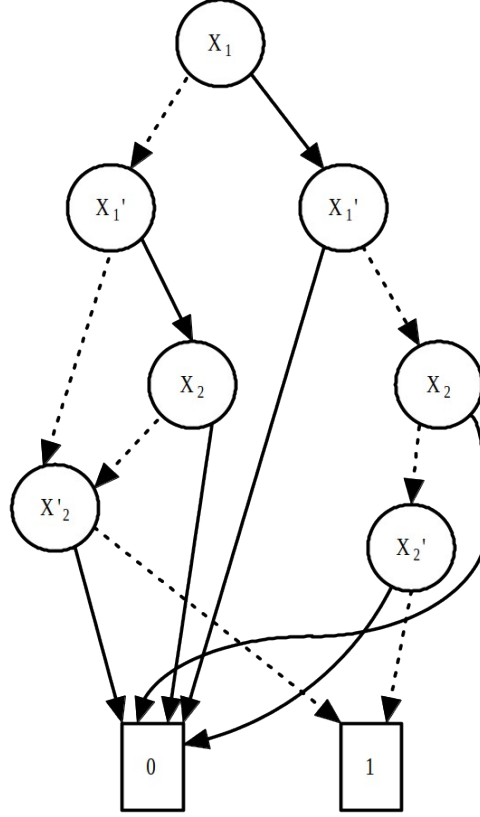


Figure 2: OBDD for Transition Relation of the model of Figure 1

Generally, in model checking algorithm, the Kripke structure M and a CTL formula Φ are given as input and the output of the algorithm is the set of states of the model which satisfy Φ . While checking for Φ , we know that the model must be labelled by the sub-formulas of Φ . Also it is clear that the primitive sub-formulas are the atomic propositions. From the labelling function of the Kripke structure, we know the states in which an atomic proposition p is true. This set of states can also be represented by an OBDD.

The basis of model checking algorithm is to find out the previous state of a given state or a set of states. Consider the labelling algorithm of $EX \Phi$. Label any state with $EX \Phi$ if one of its successors is labelled with Φ . While going to label the states with $EX \Phi$, we must label the states of the Kripke structure where Φ is true. Therefore, we know the states where Φ true, and this set of states can be represented by OBDD. The job of model checking algorithm for EX is to find the predecessor states of this state.

We have two methods used in OBDD, one is $Pre_{\exists}(X)$ and the other is $Pre_{\forall}(X)$.

$\text{Pre}_{\exists}(X)$: takes a subset X of states (of OBDD) and returns the set of states which can make a transition into X .

$\text{Pre}_{\forall}(X)$: takes a subset X of states (of OBDD) and returns the set of states which can make a transition *only* into X .

How to Find $\text{Pre}_{\exists}(X)$:

Given,

B_X : OBDD for set of all states where X is true.

$B_{\rightarrow}$: OBDD for transition relations.

Procedure,

- Rename the variables in B_X to their primed versions; call the resulting OBDD B_X' .
- Compute the OBDD for $\text{exists}(\vec{X}', \text{apply}(\bullet, B_{\rightarrow}, B_X'))$ using the **apply** and **exists** algorithms.

The working of this algorithm is explained as follows.

$\vec{X}$ is the vector of present state variables.

$\vec{X}'$ is the vector of next state variables.

Consider a set of states X and its OBDD representation is B_X . The variables involved in this OBDD are from the vector $\vec{X}$. We know that if we rename the variables of a Boolean expression, the truth value of the expression remains same. Therefore, if we rename the variables of an OBDD, the resultant OBDD still represents the same Boolean expression. If we rename the variables of B_X to get a new OBDD B_X' , then B_X' represents the same set of states like B_X , only changes are the variables which are from the vector $\vec{X}'$, that is, represented by next state variables. $B_{\rightarrow}$ is the OBDD representation of the transition system that involves variables from $\vec{X}$ and $\vec{X}'$. The OBDD $B_{\rightarrow}$ contains all the transitions of the model. Once we perform $\text{apply}(\bullet, B_{\rightarrow}, B_X')$, then it will give us those transitions which are making a transition to the set of states represented by B_X' , and the variables involved in the resultant OBDD are from $\vec{X}$ and $\vec{X}'$. The entire operation $\text{exists}(\vec{X}', \text{apply}(\bullet, B_{\rightarrow}, B_X'))$ returns an OBDD which involves variables from the vector $\vec{X}$. It makes the BDD independent of the vector $\vec{X}'$, and giving us the set of states from where those transitions are originating.

Therefore, $\text{Pre}_{\exists}(X)$ gives us the states which can make a transition into set of states X .

Now the concept will be illustrated using an example.

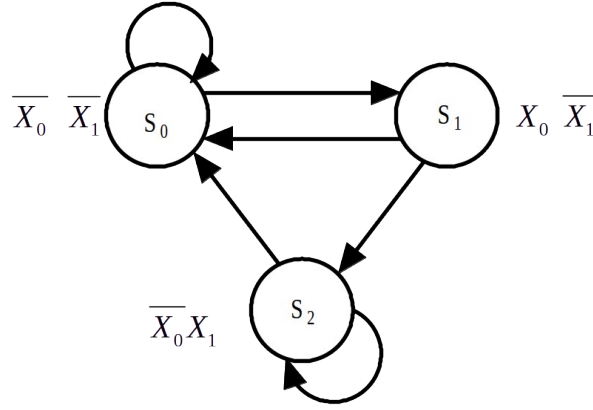


Figure 3: Example Kripke structure

Let us try to find $\text{Pre}_{\exists}(X_1)$ in the above Kripke structure of Figure 3. The atomic proposition X_1 is true in state S_2 only. Thus, B_{X_1} is the OBDD representing $\{S_2\}$. Thus B_{X_1}' (primed) is the same OBDD but with primed versions of the variables. OBDD B_{X_1}' is shown in Figure 4 and the OBDD $B_{\rightarrow}$ is shown in Figure 5.

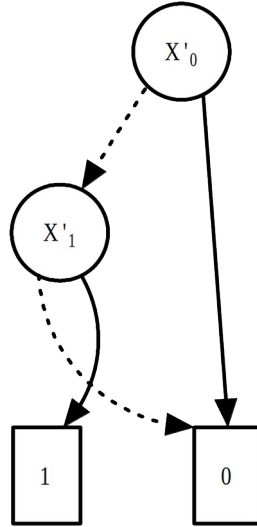


Figure 4: B_{X_1}'

The result of $\text{apply}(\bullet, B_{\rightarrow}, B_{X_1}')$ is shown in Figure 6. And the result of $\text{exists}(\bullet, \text{apply}(\bullet, B_{\rightarrow}, B_{X_1}'))$ is shown in Figure 7. Note that this OBDD is the same as OBDD for $\{S_1, S_2\}$.

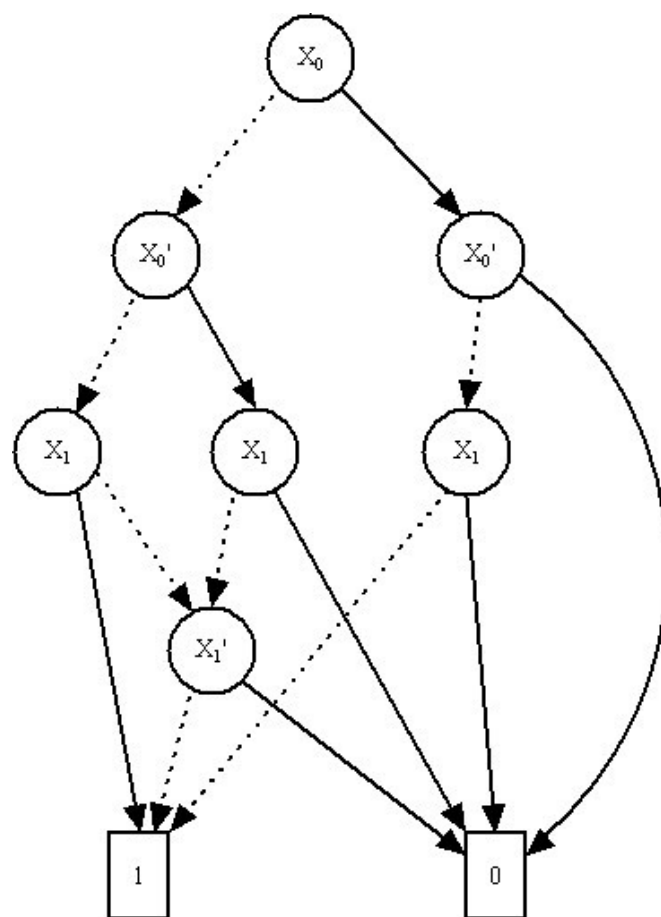


Figure 5: $B \rightarrow$

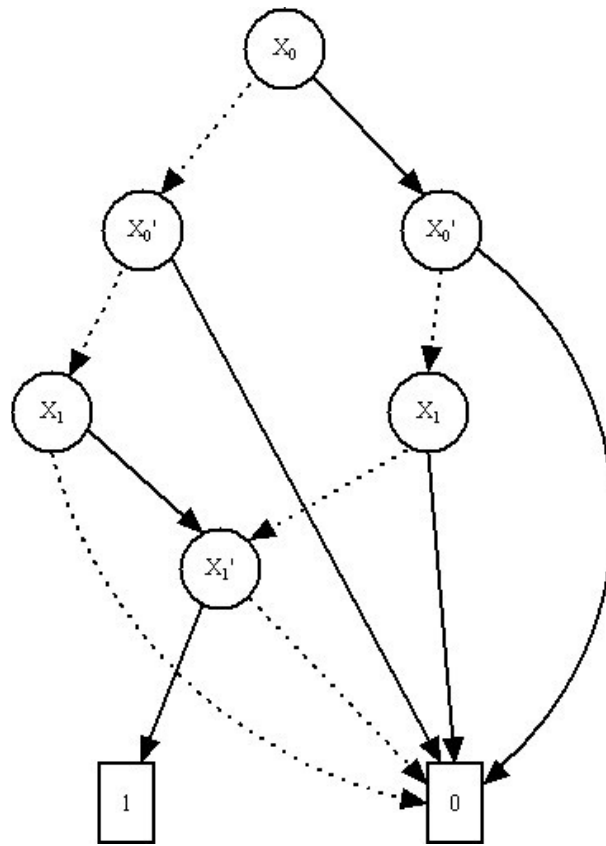


Figure 6: Result of running $\text{apply}(\bullet, B_{\rightarrow}, B_{x1'})$

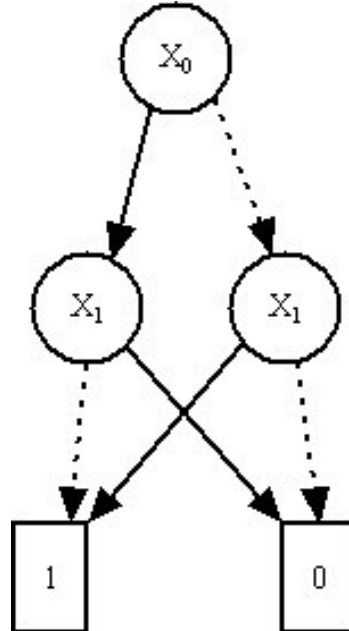


Figure 7: Result of running $\text{exists}(\hat{x}', \text{apply}(\bullet, B_{\rightarrow}, B_{x1'}))$

How to Compute $\text{Pre}_\forall(X)$:

$\text{Pre}_\forall(X)$ can be represented in terms of $\text{Pre}_\exists(X)$ as follows,

$$\text{Pre}_\forall(X) = S - \text{Pre}_\exists(S - X)$$

Consider S is the set of all the states of the transition system and X is the set of states of our interest. $(S - X)$ gives us the set of states that are not of our interest. $\text{Pre}_\exists(S - X)$ returns us the states that are having at least one transitions to the set $(S - X)$, and we are not interested for those states. The states which do not have any transition to $(S - X)$ are of our interest, and so the remaining states of S are the correct choice for $\text{Pre}_\forall(X)$.

With the help of $\text{Pre}_\exists(X)$ and $\text{Pre}_\forall(X)$ function, we can implement the symbolic model checking algorithms for operators EX, AF and EU. Now we discuss these symbolic model checking algorithms. Also analogy is pointed out with the corresponding algorithms presented in Section 2, Module 6, Lecture II and III.

EX(B_ϕ):

B_ϕ :OBDD for set of states where ϕ is true. // Analogous to $X := \text{SAT}(\psi)$; Section 2.1 Module 6, Lecture II and III.

$B_\rightarrow$:OBDD for transition relation.

Return $\text{Pre}_\exists(B_\phi)$. // Analogous to $Y := \{s \in S \mid \text{exists } s', (s \rightarrow s' \text{ and } s' \in X)\}$;

AF(B_ϕ):

B_ϕ :OBDD for set of states where ϕ is true.// Analogous to “ $Y := \text{SAT}(\psi)$ ”; Section 2.2 Module 6, Lecture II and III.

$B_\rightarrow$:OBDD for transition relation.

B_X : OBDD for all states of the system. // Analogous to “ $X=S$ ”;

repeat until $B_X=B_\phi$ // Analogous to “Repeat until $X=Y$ ”

$B_X:=B_\phi$ // Analogous to “ $X := Y$ ”;

$B_\phi:=\text{apply}(+, B_\phi, \text{Pre}_\forall(B_\phi))$ // Analogous to “ $Y:= Y \cup \{s \in S \mid \text{for all } s', (s \rightarrow s' \text{ implies } s' \in Y)\}$ ”

end

return B_ϕ

EU(B_{ψ_1}, B_{ψ_2}):

B_x : OBDD for all states of the system. // *Analogous to “ $X := S;$ ” Section 2.3 Module 6, Lecture II and III.*

B_{ψ_1} : OBDD for set of states where ψ_1 is true. // *Analogous to “ $W := SAT(\psi_1);$ ”*

B_{ψ_2} : OBDD for set of states where ψ_2 is true. // *Analogous to “ $Y := SAT(\psi_2);$ ”*

$B_{\rightarrow}$: OBDD for transition relation.

repeat until $B_x = B_{\psi_2}$

$B_x := B_{\psi_2}$ // *Analogous to “ $X := Y;$ ”*

$B_{\psi_2} := \text{apply}(+, B_{\psi_2}, \text{apply}(\bullet, B_{\psi_1}, \text{Pre}_{\exists}(B_{\psi_2})))$ // *Analogous to “ $Y := Y \cup (W \cap \{ s \in S \mid \text{exists } s', (s \rightarrow s' \text{ and } s' \in Y) \});$ ”*

end

return B_{ψ_2}

3. Examples of calculating EX, AF and AU using symbolic model checking

In this section we present some examples of symbolic model checking using the Kripke structure of Figure 8. The state encoding in terms of X_1 and X_2 are shown in the same figure.

OBDDs for different subsets of states of the Kripke structure of Figure 8 are shown in Figure 9 through Figure 14.

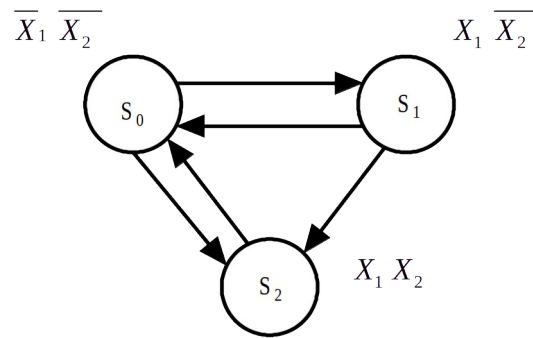


Figure 8. Example of Kripke Structure

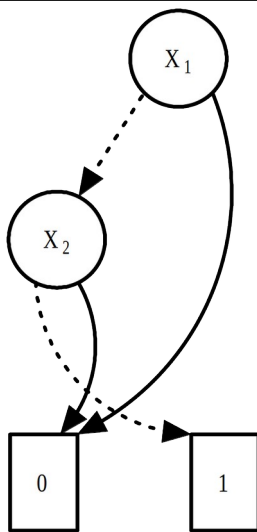


Figure 9. OBDD for $\{S_0\}$

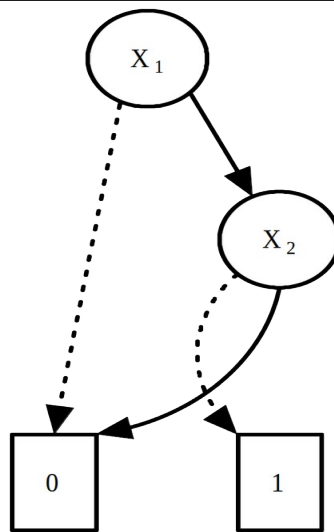


Figure 10. OBDD for $\{S_1\}$

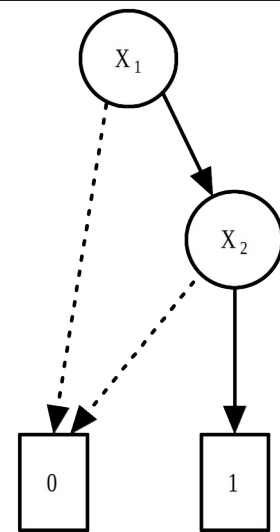


Figure 11. OBDD for $\{S_2\}$

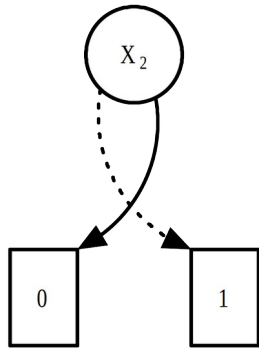


Figure12. OBDD for $\{S_0, S_1\}$

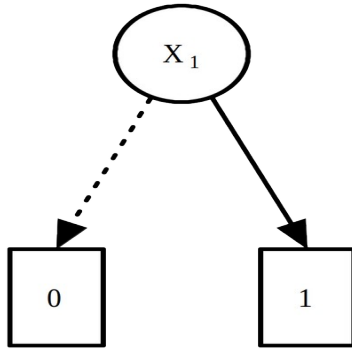


Figure13. OBDD for $\{S_1, S_2\}$

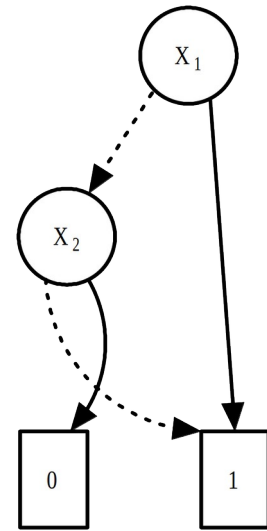


Figure14. OBDD for $\{S_0, S_1, S_2\}$

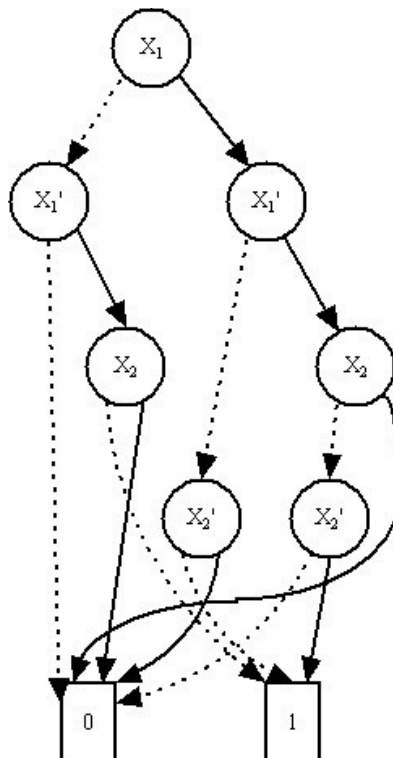


Figure15. OBDD for Transition Relation ($B \rightarrow$)

The OBDD representation of the transition relation of Kripke structure of Figure 8 is given in Figure 15. In the following examples, we will demonstrate how the symbolic model checking algorithm is used to find the set of states where a given formula is true.

- **EX(B_φ)** where $\varphi = X_1$

B_φ = OBDD of set of states where X_1 is true = OBDD for $\{S_1, S_2\}$.

$B_\varphi' = B_\varphi$, but all variables are renamed to their primed versions.

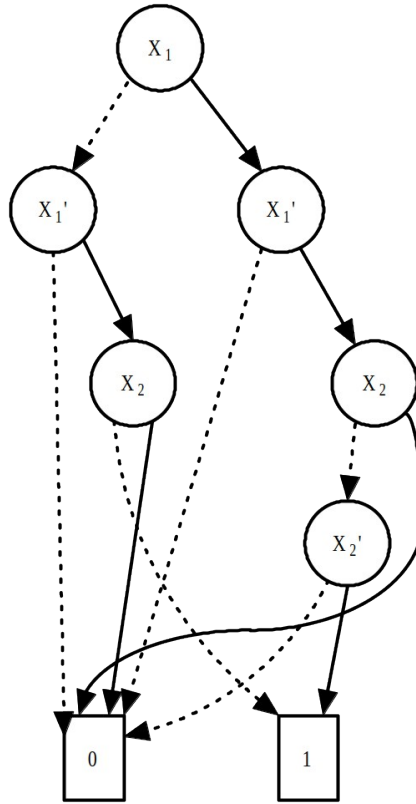


Figure16: Resultant OBDD of $\text{apply}(\bullet, B_\rightarrow, B_\varphi')$

Figure 16 shows the resultant OBDD after performing the operation $\text{apply}(\bullet, B_\rightarrow, B_\varphi')$.

Applying $\text{exists}(\vec{X}', \text{apply}(\bullet, B_\rightarrow, B_\varphi'))$ yields the OBDD as shown in Figure 17. This OBDD is the same as the OBDD for states $\{S_0, S_1\}$. Thus S_0 and S_1 are the states where $\text{EX}(X_1)$ holds.

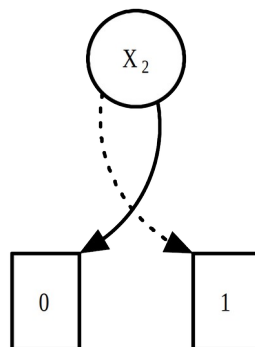


Figure17: Result of $\text{EX}(B_\varphi)$

- **AF(B_φ)** where $\varphi = X_1$:

1. B_φ = OBDD of set of states where X_1 is true = OBDD for $\{S_1, S_2\}$; Figure 18

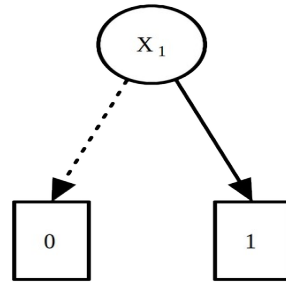


Figure 18. OBDD for B_φ

2. For calculation of $\text{Pre}_\forall(B_\varphi)$

- $B_{S-\varphi}$ = OBDD of set of states where X_1 is not true = OBDD for $\{S_0\}$; Figure 19.

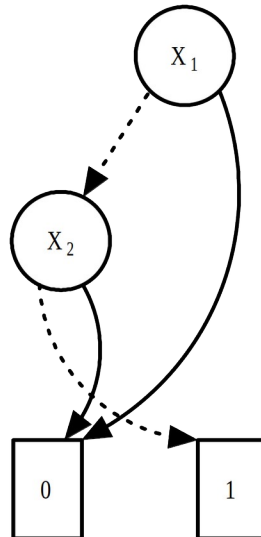


Figure 19. OBDD for $B_{S-\varphi}$,

- $\text{Pre}_\exists(B_{S-\varphi})$ = OBDD of $\{S_1, S_2\}$; Figure 20.

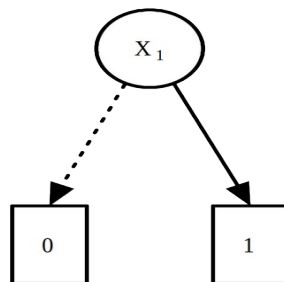


Figure 20. OBDD of $\text{Pre}_\exists(B_{S-\varphi})$

- $S \vdash \text{Pre}\exists(B_S \neg \varphi)$, i.e., complementing $\{S_1, S_2\}$ with respect to S we get $\{S_0\}$, thus $\text{Pre}\forall(B_\varphi) = \{S_0\}$; Figure 19.

3. $\text{apply}(+, B_\varphi, \text{Pre}\forall(B_\varphi))$ is $\text{apply}(+, \text{OBDD of } \{S_1, S_2\}, \text{OBDD of } \{S_0\}) = \text{OBDD of } \{S_1, S_2, S_0\}$; Figure 21. Since all the states are already included; there is no point in continuing the iteration. Thus the algorithm stops here.

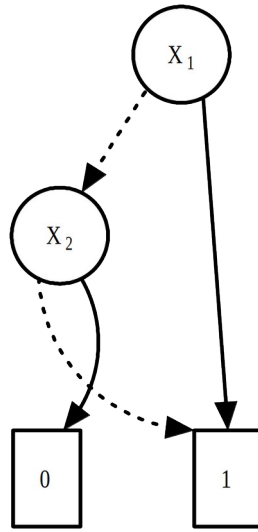


Figure 21. Result of $\text{AF}(B_\varphi)$, which represent OBDD for all states

4. Conclusions

In this lecture we have discussed about symbolic model checking, where we use OBDD to represent the state transition system and the set of states. So OBDD manipulation algorithms are used to find the set of states where a given CTL formula is true. Since with proper variable ordering most of the time we get a compact OBDD representation of a system, therefore, it becomes possible to apply model checking for bigger and complex systems.

In the second module of the course on “Verification” we mainly covered concepts to design a verifiable model a system and then verify if all the design intents are met by the model. As design phase involves many transformations, so verifying that all intermediate design forms are equivalent and satisfy the design specifications is mandatory. Verification is the technique to ensure this fact. Once a design is verified, it is sent for fabrication. Fabrication step is a manufacturing process and may have inaccuracies which leads to faults in the circuit, e.g., output of a gate may always give logic 1 as output, irrespective of the inputs. To ensure that only fault free chips are delivered, we need to test all the manufactured chips. The next part of the course on “Testing” discusses in details the VLSI testing flow.

Question and Answers

Question:

In the example of Figure 8, using symbolic model checking determine the states where $EU(B_{\psi_1}, B_{\psi_2})$ where $\psi_1 = X_1$ and $\psi_2 = X_2$

Answer:

Iteration 1

1. OBDD of $\psi_1 = X_1$, states $\{S_1, S_2\}$; Figure 22
2. OBDD of $\psi_2 = X_2$, states $\{S_2\}$; Figure 23

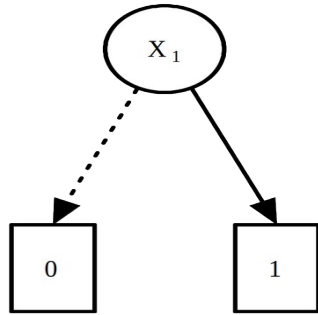


Figure 22. B_{ψ_1}

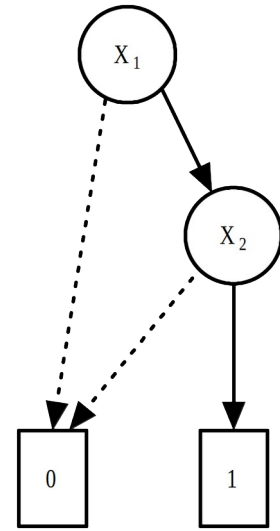


Figure 23. B_{ψ_2}

3. $\text{Pre}_{\exists}(B_{\psi_2}) = \text{OBDD of states } \{S_0, S_1\}$; Figure 24
4. Applying $(.)$ operation on B_{ψ_1} and $\text{Pre}_{\exists}(B_{\psi_2}) = \text{OBDD of } \{S_1\}$; Figure 25

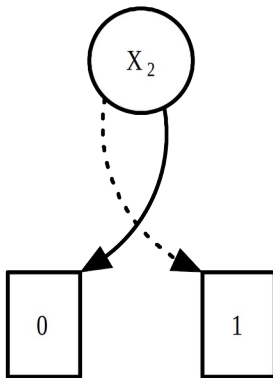


Figure 24. $\text{Pre}_{\exists}(B_{\psi_2})$

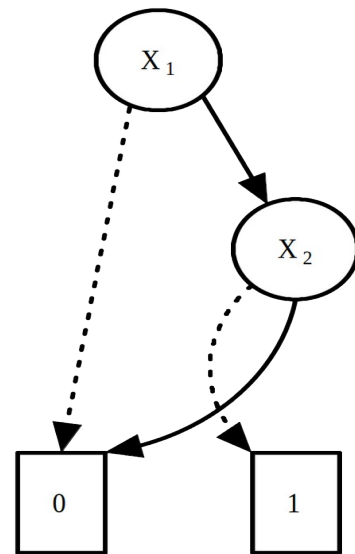


Figure 25. Applying $(.)$ operation on B_{ψ_1} and $\text{Pre}_{\exists}(B_{\psi_2})$

5. Applying (+) operation on B_{ψ_2} and OBDD of $\{S_1\}$ (from Step 4)= OBDD of $\{S_1, S_2\}$; Figure 26.

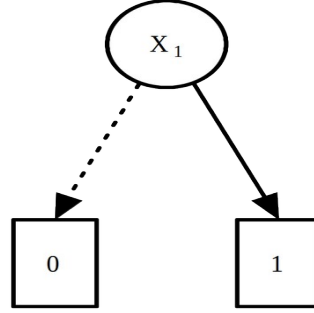


Figure 26. Applying (+) operation on B_{ψ_2} and $\text{apply}(\bullet, B_{\psi_1}, \text{Pre}_{\exists}(B_{\psi_2}))$

Second Iteration

1. OBDD of $\psi_1 = X_1$, states $\{S_1, S_2\}$; Figure 22
2. OBDD of $\psi_2 =$ states $\{S_1, S_2\}$; Figure 26
3. $\text{Pre}_{\exists}(B_{\psi_2}) =$ OBDD of states $\{S_0, S_1\}$; Figure 27

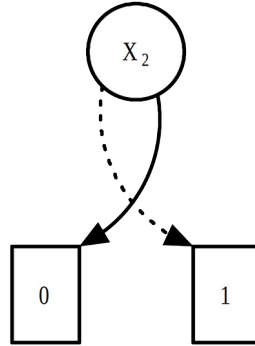


Figure 27. Result of $\text{Pre}_{\exists}(B_{\psi_2})$

4. Applying (.) operation on B_{ψ_1} and $\text{Pre}_{\exists}(B_{\psi_2}) =$ OBDD of $\{S_1\}$; Figure 28

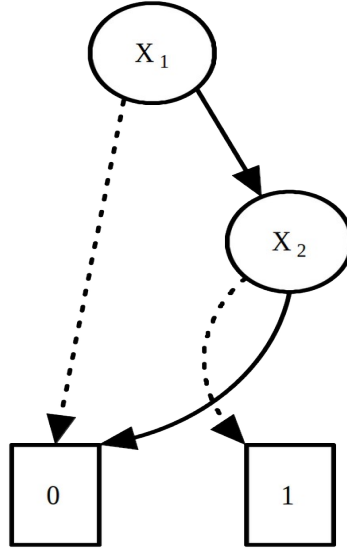


Figure 28. Applying (.) operation on $B_{\psi1}$ and $\text{Pre}_{\exists}(B_{\psi2})$

5. Applying (+) operation on $B_{\psi2}$ and OBDD of $\{S_1\}$ (from Step 4)= OBDD of $\{S_1, S_2\}$; Figure 29.

There is no change in $B_{\psi2}$ after this iteration. So algorithm stops. Final $B_{\psi2}$ represents $\{S_1, S_2\}$ meaning these are the states where $\text{EU}(B_{\psi1}, B_{\psi2})$ is true.

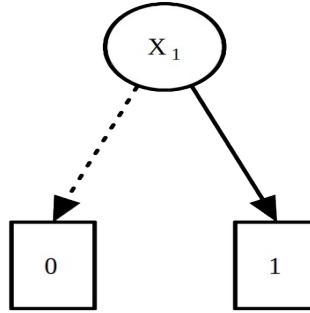


Figure 29. Applying (+) operation on $B_{\psi2}$ and $\text{apply}(\bullet, B_{\psi1}, \text{Pre}_{\exists}(B_{\psi2}))$